

Ke Zhang

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EDUCATION

University of California, Riverside — Ph.D. in Mathematics, Expected May 2028

Northeastern University — M.S. in Mathematics, May 2023

Southwest Minzu University — B.Econ. in Financial Engineering, June 2021

SELECTED RESEARCH

Beyond Compilation: Evaluating Faithful Natural-Language-to-Lean Statement Formalization

- Measured the compilation–faithfulness gap for a tool-augmented Lean formalization agent on a 400-statement benchmark of open-ended, graduate-level mathematics, showing that compile rate can substantially overstate formalization quality.
- Used a controlled 2^3 factorial study to isolate how expert drafting, Mathlib search, and Lean feedback affect validity and faithfulness.

PHREEQC-MCQ-200: A Diagnostic Benchmark for Tool-Augmented Scientific Simulator Agents

- Built PHREEQC-MCQ-200, a 200-question benchmark and sandboxed agent framework for evaluating LLM agents on PHREEQC, a deterministic aqueous-geochemistry simulator.
- Introduced retention diagnostics showing that tool use can improve aggregate accuracy while breaking previously correct items; compared raw-output and TOC-guided access, finding that interface effectiveness varies with model capability.

Learning Parameterized Nonlinear Elasticity on Curved Surfaces

- Led PyTorch training of a parameter-conditioned PINN, achieving $\sim 10^{-3}$ loss and generalization to unseen parameter combinations.
- Conducted architecture and collocation-density ablations to identify stable configurations and failure modes.

Agent-Based Code Repair for Gurobi Optimization Models

- Built an agentic Gurobi code-repair system and a 26-problem, 260-test benchmark with controlled bug generation and API-grounded retrieval. Used factorial experiments to quantify how tool combinations affect repair performance.

TEACHING

Teaching Assistant, UC Riverside, 2024–2026 — MATH 004L, 005A, 006A/B, 007A/B, 009A/B, 120

Teaching Assistant, Northeastern University, 2022–2023 — Group Theory

TECHNICAL SKILLS

Programming and Tools: Python, C/C++, PyTorch, Lean 4, Mathlib, Docker, PHREEQC, Gurobi/Gurobipy, LaTeX

Methods: AI agents, formal reasoning, AI4Math, AI4Science, benchmark and evaluation design

INTERESTS

Mountaineering & endurance: Summited Mt. Siguniang Erfeng (5,276 m, 17309 ft), Sichuan, China; completed Mt. San Jacinto via Devil's Slide Trail (16 mi) Palm Spring, CA.

Hands-on mechanical work: Perform own vehicle maintenance, including oil changes and brake pad/rotor replacement.